Implement a Generative Adversarial Network (GAN) to generate synthetic images or tabular data

Aim:

To implement a Generative Adversarial Network (GAN) using the MNIST dataset to learn the data distribution of handwritten digit images. To train the generator and discriminator models so that the generator can produce realistic synthetic digit images.

Dataset : MNIST

Step 1 – Import Required Libraries

```
import torch

import torch.nn as nn

import torchvision

import torchvision.transforms as transforms

import matplotlib.pyplot as plt
```

Step 2 – Load MNIST Dataset

```
transform = transforms.Compose([

   transforms.ToTensor(),

   transforms.Normalize((0.5,), (0.5,))

])

dataset = torchvision.datasets.MNIST(

   root='./data',

   train=True,

   download=True,

   transform=transform

)

dataloader = torch.utils.data.DataLoader(

   dataset,

   batch_size=64,

   shuffle=True )
```

```
100%|██████████| 9.91M/9.91M [00:00<00:00, 66.8MB/s]
100%|██████████| 28.9k/28.9k [00:00<00:00, 1.74MB/s]
100%|██████████| 1.65M/1.65M [00:00<00:00, 15.3MB/s]
100%|██████████| 4.54k/4.54k [00:00<00:00, 11.4MB/s]
```

Step 3 – Create Generator Model

```
generator = nn.Sequential(
    nn.Linear(100,256),
    nn.ReLU(),
    nn.Linear(256,512),
    nn.ReLU(),
    nn.Linear(512,1024),
    nn.ReLU(),
    nn.Linear(1024,784),
    nn.Tanh()
)
```

Step 4 – Create Discriminator Model

```
discriminator = nn.Sequential(
    nn.Linear(784,512),
    nn.LeakyReLU(0.2),
    nn.Linear(512,256),
    nn.LeakyReLU(0.2),
    nn.Linear(256,1),
    nn.Sigmoid()
)
```

Step 5 – Define Loss Function

```
loss_function = nn.BCELoss()
```

Step 6 – Define Optimizers

```
optimizer_generator = torch.optim.Adam(
    generator.parameters(),
    lr=0.0002
)
optimizer_discriminator = torch.optim.Adam(
    discriminator.parameters(),
    lr=0.0002
)
```

Step 7 – Train the GAN

```
epochs = 10
for epoch in range(epochs):
    for images,labels in dataloader:
        real_images = images.view(-1,784)
        real_labels = torch.ones(real_images.size(0),1)
        fake_labels = torch.zeros(real_images.size(0),1)
        outputs = discriminator(real_images)
        loss_real = loss_function(outputs,real_labels)
        noise = torch.randn(real_images.size(0),100)
        fake_images = generator(noise)
        outputs = discriminator(fake_images.detach())
        loss_fake = loss_function(outputs,fake_labels)
        loss_discriminator = loss_real + loss_fake
        optimizer_discriminator.zero_grad()
        loss_discriminator.backward()
        optimizer_discriminator.step()
        noise = torch.randn(real_images.size(0),100)
```

```
    fake_images = generator(noise)
    outputs = discriminator(fake_images)
    loss_generator = loss_function(outputs,real_labels)
    optimizer_generator.zero_grad()
    loss_generator.backward()
    optimizer_generator.step()

  print("Epoch:",epoch,"Generator Loss:",loss_generator.item(),"Discriminator
Loss:",loss_discriminator.item())
```

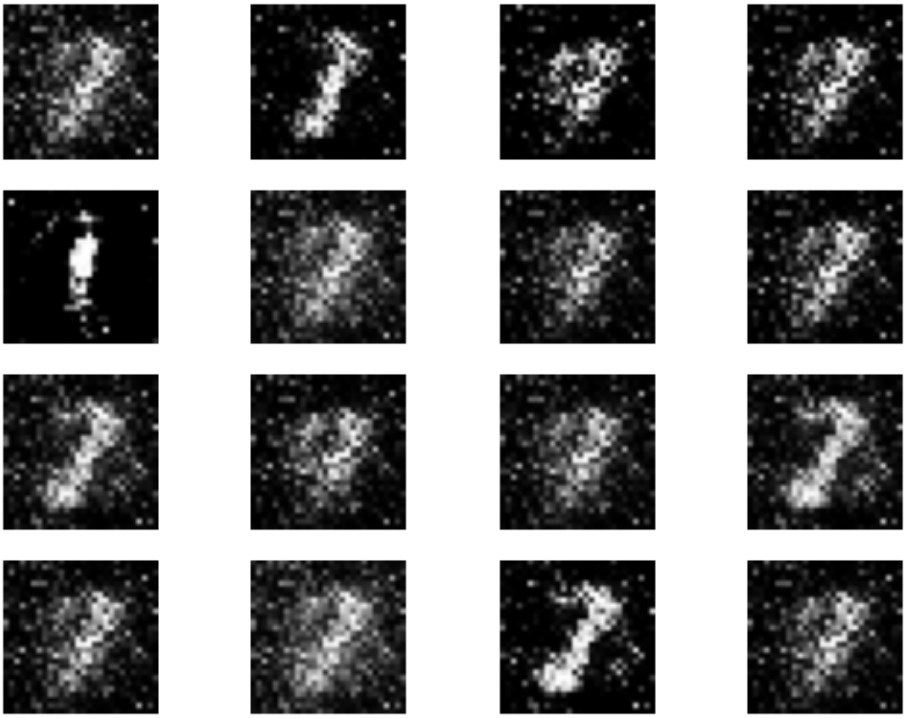
```
Epoch: 0 Generator Loss: 5.954799175262451 Discriminator Loss: 0.05623356252908707
Epoch: 1 Generator Loss: 4.196893692016602 Discriminator Loss: 1.1113841533660889
Epoch: 2 Generator Loss: 2.1464388370513916 Discriminator Loss: 0.7769961357116699
Epoch: 3 Generator Loss: 3.4867875576019287 Discriminator Loss: 0.40428319573402405
Epoch: 4 Generator Loss: 3.1963438987731934 Discriminator Loss: 0.2141532599925995
Epoch: 5 Generator Loss: 3.7585525512695312 Discriminator Loss: 0.4819808006286621
Epoch: 6 Generator Loss: 5.783203125 Discriminator Loss: 0.5697464942932129
Epoch: 7 Generator Loss: 3.208303451538086 Discriminator Loss: 0.3794306218624115
Epoch: 8 Generator Loss: 3.0558903217315674 Discriminator Loss: 0.7844176292419434
Epoch: 9 Generator Loss: 2.43078875541687 Discriminator Loss: 0.7860702276229858
```

Step 8 – Generate Synthetic Images

```
noise = torch.randn(16,100)
generated_images = generator(noise)
generated_images = generated_images.view(16,28,28)
generated_images = generated_images.detach().numpy()
```

Step 9 – Display Generated Images

```
for i in range(16):
   plt.subplot(4,4,i+1)
   plt.imshow(generated_images[i],cmap='gray')
   plt.axis('off')
plt.show()
```



Result:

The GAN model was successfully trained on the MNIST dataset and generated synthetic handwritten digit images. The generated images resemble MNIST digit patterns, demonstrating that the model learned the underlying data distribution.